CRITICAL THINKING In Exercises 56–63, complete the statement with *always*, *sometimes*, or *never*. Explain your reasoning.

56. A line never has endpoints.

57. A line and a point always intersect.

58. A plane and a point always intersect.

59. Two planes always intersect in a line.

60. Two points always determine a line.

61. Any three points sometimes determine a plane.

62. Any three points not on the same line always determine a plane.

63. Two lines that are not parallel always intersect.

In Exercises 39–44, use the diagram to name all the points that are not coplanar with the given points.

39. N, K, and L R, Q, P & S

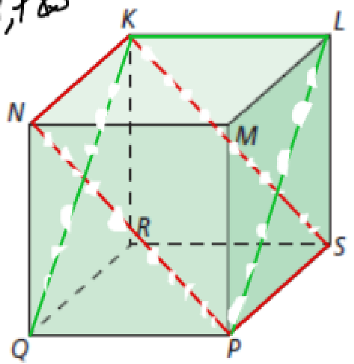
40. P, Q, and N

41. P, Q, and R

42. R, K, and N

43. P, S, and K

44. Q, K, and L



40) R, K, L & S

41) M, N, K & L

42) M, P, S & L

43) Q, L, M & R

44) N, R, M & S

Using the Ruler Postulate

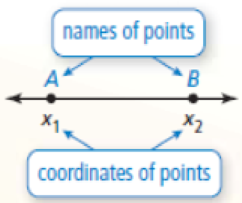
In geometry, a rule that is accepted without proof is called a **postulate** or an **axiom**. A rule that can be proved is called a *theorem*, as you will see later. Postulate 1.1 shows how to find the distance between two points on a line.

$12 - (-5) = 12 + 5 = 17$

Postulate

Postulate 1.1 Ruler Postulate

The points on a line can be matched one to one with the real numbers. The real number that corresponds to a point is the **coordinate** of the point.



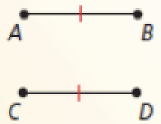
The **distance** between points A and B, written as AB, is the absolute value of the difference of the coordinates of A and B.

$AB = |x_2 - x_1|$

Core Concept

Congruent Segments

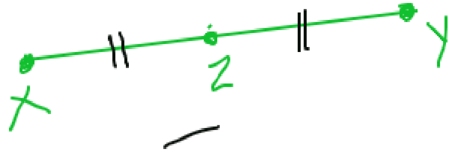
Line segments that have the same length are called **congruent segments**. You can say "the length of $\overline{AB}$ is equal to the length of $\overline{CD}$," or you can say "$\overline{AB}$ *is congruent to* $\overline{CD}$." The symbol $\cong$ means "is congruent to."



Lengths are equal.
$AB = CD$
"is equal to"

OR

Segments are congruent.
$\overline{AB} \cong \overline{CD}$
"is congruent to"



$\overline{XZ} \cong \overline{YZ}$

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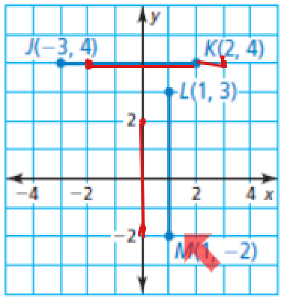
READING

In the diagram, the red tick marks indicate $\overline{AB} \cong \overline{CD}$. When there is more than one pair of congruent segments, use multiple tick marks.

EXAMPLE 2 Comparing Segments for Congruence

Plot $J(-3, 4)$, $K(2, 4)$, $L(1, 3)$, and $M(1, -2)$ in a coordinate plane. Then determine whether $\overline{JK}$ and $\overline{LM}$ are congruent.

SOLUTION



Plot the points, as shown. To find the length of a horizontal segment, find the absolute value of the difference of the *x*-coordinates of the endpoints.

✓ $JK = |2 - (-3)| = 5$ Ruler Postulate

To find the length of a vertical segment, find the absolute value of the difference of the *y*-coordinates of the endpoints.

✓ $LM = |-2 - 3| = 5$ Ruler Postulate

▶ $\overline{JK}$ and $\overline{LM}$ have the same length. So, $\overline{JK} \cong \overline{LM}$.

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5. Plot $A(-2, 4)$, $B(3, 4)$, $C(0, 2)$, and $D(0, -2)$ in a coordinate plane. Then determine whether $\overline{AB}$ and $\overline{CD}$ are congruent.

$AB = |2-(-3)| = 5$

$CD = |-2-2| = 4$

$\overline{AB} \not\cong \overline{CD}$

Using the Segment Addition Postulate

When three points are collinear, you can say that one point is **between** the other two.

Point B is between points A and C.

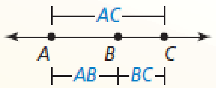
Point E is not between points D and F.

Postulate

Postulate 1.2 Segment Addition Postulate

If B is between A and C, then $AB + BC = AC$.

If $AB + BC = AC$, then B is between A and C.



$DF = DE + EF = 58$

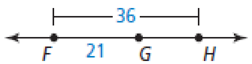
EXAMPLE 3 Using the Segment Addition Postulate

a. Find DF.

58 units

$GH = ?$

b. Find GH.



$FG + GH = FH$

$21 + GH = 36$

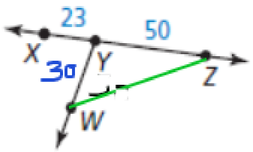
$GH = 36 - 21 = 15$ units

Q) Use the diagram at the right.

73 units

6. Use the Segment Addition Postulate to find XZ.

7. In the diagram, $WY = 30$. Can you use the Segment Addition Postulate to find the distance between points W and Z? Explain your reasoning.



No.

The points W, Y, and Z have to be collinear in order to apply segment addition postulate.

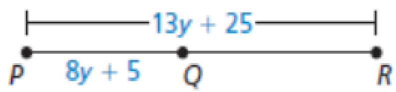
MATHEMATICAL CONNECTIONS Write an expression for the length of the segment.

a. $\overline{AC}$ ✓

$AC = (x+2) + (7x-3)$

$8x - 1$

b. $\overline{QR}$



$QR = (13y+25) - (8y+5)$

$= (5y+20)$